

1. How to Check the List of Running Processes?

```
sneha_mhasalkar@cloudshell:~/day2$ ps
  PID TTY          TIME CMD
   698 pts/2        00:00:00 bash
  2259 pts/2        00:00:00 ps
sneha_mhasalkar@cloudshell:~/day2$
```

2. How to print PID of the current shell?

```
sneha_mhasalkar@cloudshell:~/day2$ echo "the process id PID : $$"
the process id PID : 699
sneha_mhasalkar@cloudshell:~/day2$
```

3. Write a Shell Script that adds two numbers if provided as the command Line Argument and if the two numbers are not entered it outputs an Error Message.

```
sneha_mhasalkar@cloudshell:~/day2$ ./que2.sh
Error: Please provide two numbers as arguments.
sneha_mhasalkar@cloudshell:~/day2$
sneha_mhasalkar@cloudshell:~/day2$ ./que2.sh 1 2
res: 3
sneha_mhasalkar@cloudshell:~/day2$
```

4. Create a Script test1.sh which will be executed with arguments: 10 20 Virat Kohli "Rohit Sharma"

e.g. sh test1.sh 10 20 Kamini Patil "Kamini Patil"

Use \$* and \$@ to print the command line arguments in the format below:

```
10
20
Virat
Kohli
Rohit
Sharma
```

```
10
20
Virat
```

Kohli
Rohit Sharma

```
sneha_mhasalkar@cloudshell:~/day2$ vi que4.sh
sneha_mhasalkar@cloudshell:~/day2$ ./que4.sh 10 20 Virat Kohli "Rohit Sharma"
by using *
10
20
Virat
Kohli
Rohit
Sharma
by using @
10
20
Virat
Kohli
Rohit Sharma
sneha_mhasalkar@cloudshell:~/day2$ █
```

5. Write a script to check if the commands below are successful or not in Unix?
- ls -l <filename that does not exist in the current directory>
 - ls -l <filename that exist in the current directory>

What was the status return when the last command executed was successful and when the last command executed was not successful.

```
sneha_mhasalkar@cloudshell:~/day2$ ./que5.sh
Checking a file that does not exist
ls: cannot access 'file1.txt': No such file or directory
Command failed
Checking a file that exists
-rwxrwxr-x 1 sneha_mhasalkar sneha_mhasalkar 318 Jun 22 04:33 que5.txt
Command executed successfully
sneha_mhasalkar@cloudshell:~/day2$ █
```

6. Write a script to create an array of 10 for storing Employee name and print all array elements and their respective indexes?

```
sneha_mhasalkar@cloudshell:~/day2$ vi que6.sh
sneha_mhasalkar@cloudshell:~/day2$ ./que6.sh
jk rm jimin jin v suga hope kim park hwang
index : 0 = jk
index : 1 = rm
index : 2 = jimin
index : 3 = jin
index : 4 = v
index : 5 = suga
index : 6 = hope
index : 7 = kim
index : 8 = park
index : 9 = hwang
sneha_mhasalkar@cloudshell:~/day2$ █
```

7. Accept the index number to be deleted from the array created in the previous example as command line argument to the script, Remove the elements and display the entire array again.

```
sneha_mhasalkar@cloudshell:~/day2$ vi que7.sh
sneha_mhasalkar@cloudshell:~/day2$ ./que7.sh 2
before deleting : jk rm jimin jin v suga hope kim park hwang
index : 0 = jk
index : 1 = rm
index : 3 = jin
index : 4 = v
index : 5 = suga
index : 6 = hope
index : 7 = kim
index : 8 = park
index : 9 = hwang
sneha_mhasalkar@cloudshell:~/day2$ █
```

8. Write 5 functions below:
 - a. addition() # Function to add 2 numbers
 - b. subtraction() #Function to subtract second number from first number
 - c. multiplication() #Function to multiply 2 number
 - d. division() #Function to divide second number by first number
 - e. Write a function to display a menu as below,


```
***** calculator *****
```

0. exit"
1. addition"
2. subtraction"
3. multiplication"
4. division"

Enter the choice :

Ask the user to enter the choice and then 2 numbers to perform the calculation as per the choice provided by the user. If the user enters 0 as choince display the message "Closing the calculator"

To solve the problem i have use the switch case and take the user input for the choice as well as the input for perform the operation

```
sneha_mhasalkar@cloudshell:~/day2$ vi que8.sh
sneha_mhasalkar@cloudshell:~/day2$ ./que8.sh
enter the input to perfrom operation
Enter first number (a):
2
Enter second number (b):
2
--- Calculator Menu ---
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice):
1
Addition result: 4
sneha_mhasalkar@cloudshell:~/day2$
```

```
sneha_mhasalkar@cloudshell:~/day2$ vi que8.sh
sneha_mhasalkar@cloudshell:~/day2$ ./que8.sh
enter the input to perfrom operation
Enter first number (a):
5
Enter second number (b):
4
--- Calculator Menu ---
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
Enter your choice):
5
Closing the caluclatort
sneha_mhasalkar@cloudshell:~/day2$
```

9. Create a text file employee.txt as shown in the presentation today.

a. Write an awk command to print the last line.

```
sneha_mhasalkar@cloudshell:~/day2$ awk 'END {print}' que9.sh
6          jk          account          5400
sneha_mhasalkar@cloudshell:~/day2$
```

b. Write an awk command to print rows with department “account” only.

```
sneha_mhasalkar@cloudshell:~/day2$ awk '/account/ {print}' que9.sh
3          varun  manager account          120000
6          jk     account          5400
echo "row with department account"
sneha_mhasalkar@cloudshell:~/day2$
```

c. Write an awk command to print columns below:

serial number, employee name, employee salary

```
sneha_mhasalkar@cloudshell:~/day2$ awk 'NR>3 {print $1,$2,$NF}' que9.sh
1 ajay 50000
2 amit 20000
3 varun 120000
4 sunil 40000
5 tarun 2200
6 jk 5400
sneha_mhasalkar@cloudshell:~/day2$
```

- d. Do you know any alternatives to the awk commands used in a, b and c above ? If Yes, please list the commands.

10. Write a script to accept year from user and check whether leap year or not.

```
sneha_mhasalkar@cloudshell:~/day2$ ./que10.sh 2024
year is leap year
sneha_mhasalkar@cloudshell:~/day2$ ./que10.sh 2025
year is not leap year
sneha_mhasalkar@cloudshell:~/day2$ █
```

11. Write a shell script to accept the name from the user and check whether user entered name is file or directory. If name is file display its size and if it is directory display its contents.

If the user input is directory

For both i have take command line argument

```
sneha_mhasalkar@cloudshell:~/day2$ ./q11.sh demo
this is the directory
de/ demo2/ demo3/ demo4.txt
sneha_mhasalkar@cloudshell:~/day2$ █
```

If the user input is file

```
sneha_mhasalkar@cloudshell:~/day2$ ./q11.sh file.txt
this is file
Size: 0 (Total details: -rw-rw-r-- 1 sneha_mhasalkar sneha_mhasalkar 0 Jun
19 11:40 file.txt)
sneha_mhasalkar@cloudshell:~/day2$ █
```

12. Write a Program to find the greatest of three numbers

I have used Nested loop to solve this problem

And given command line argument a,b,c

```
#!/bin/bash
#
a=$1
b=$2
c=$3

if((a > b && a > c));then
    echo "a is greater than both "
elif ((b > a && b > c));then
    echo "bb is greater "
else
    echo "c is greater"
fi
~
~
~
~
~
```

```
sneha_mhasalkar@cloudshell:~/day2$ ./ques11.sh 10 2 30
c is greater
sneha_mhasalkar@cloudshell:~/day2$ █
```

13. Write a Program to find whether a given number is positive or negative

I have used if statement to solve the problem

```
sneha_mhasalkar@cloudshell:~/day2$ ./que13.sh -1
the no is negative
sneha_mhasalkar@cloudshell:~/day2$ ./que13.sh 10
no is positive
sneha_mhasalkar@cloudshell:~/day2$ █
```

```
#!/bin/bash
#
a=$1

if ((a < 0));then
    echo "the no is negative"
else
    echo "no is positive"
fi
█
~
~
```

14. Commands for below:

1. List the directory contents date wise sorted.

```
sneha_mhasalkar@cloudshell:~/day2$ ls -ltr
total 72
-rwxrwxr-x 1 sneha_mhasalkar sneha_mhasalkar 120 Jun 19 04:09 code.sh
-rwxrwxr-x 1 sneha_mhasalkar sneha_mhasalkar 128 Jun 19 05:19 p2.sh
-rwxrwxr-x 1 sneha_mhasalkar sneha_mhasalkar 94 Jun 19 05:25 p3.sh
-rwxrwxr-x 1 sneha_mhasalkar sneha_mhasalkar 70 Jun 19 05:28 funfromanother.sh
-rwxrwxr-x 1 sneha_mhasalkar sneha_mhasalkar 581 Jun 19 05:54 array.sh
-rwxrwxr-x 1 sneha_mhasalkar sneha_mhasalkar 307 Jun 19 06:27 que2.sh
-rwxrwxr-x 1 sneha_mhasalkar sneha_mhasalkar 9 Jun 19 06:32 practice.sh
-rwxrwxr-x 1 sneha_mhasalkar sneha_mhasalkar 122 Jun 19 06:37 que4.sh
-rwxrwxr-x 1 sneha_mhasalkar sneha_mhasalkar 158 Jun 19 06:56 que6.sh
-rwxrwxr-x 1 sneha_mhasalkar sneha_mhasalkar 196 Jun 19 07:01 que7.sh
-rwxrwxr-x 1 sneha_mhasalkar sneha_mhasalkar 959 Jun 19 07:21 que8.sh
-rwxrwxr-x 1 sneha_mhasalkar sneha_mhasalkar 272 Jun 19 07:40 que9.sh
-rwxrwxr-x 1 sneha_mhasalkar sneha_mhasalkar 158 Jun 19 08:09 que10.sh
-rw-rw-r-- 1 sneha_mhasalkar sneha_mhasalkar 23 Jun 19 08:35 p1.sh
-rwxrwxr-x 1 sneha_mhasalkar sneha_mhasalkar 178 Jun 19 11:30 ques11.sh
-rwxrwxr-x 1 sneha_mhasalkar sneha_mhasalkar 104 Jun 19 11:35 que13.sh
-rw-rw-r-- 1 sneha_mhasalkar sneha_mhasalkar 0 Jun 19 11:40 file.txt
-rwxrwxr-x 1 sneha_mhasalkar sneha_mhasalkar 470 Jun 19 11:48 q11.sh
drwxrwxr-x 5 sneha_mhasalkar sneha_mhasalkar 4096 Jun 19 11:49 demo
```

2. List the directory contents size wise sorted.

```
sneha_mhasalkar@cloudshell:~/day2$ ls -lhs
total 72K
4.0K -rwxrwxr-x 1 sneha_mhasalkar sneha_mhasalkar 581 Jun 19 05:54 array.sh
4.0K -rwxrwxr-x 1 sneha_mhasalkar sneha_mhasalkar 120 Jun 19 04:09 code.sh
4.0K drwxrwxr-x 5 sneha_mhasalkar sneha_mhasalkar 4.0K Jun 19 11:49 demo
0 -rw-rw-r-- 1 sneha_mhasalkar sneha_mhasalkar 0 Jun 19 11:40 file.txt
4.0K -rwxrwxr-x 1 sneha_mhasalkar sneha_mhasalkar 70 Jun 19 05:28 funfromanother.sh
4.0K -rw-rw-r-- 1 sneha_mhasalkar sneha_mhasalkar 23 Jun 19 08:35 p1.sh
4.0K -rwxrwxr-x 1 sneha_mhasalkar sneha_mhasalkar 128 Jun 19 05:19 p2.sh
4.0K -rwxrwxr-x 1 sneha_mhasalkar sneha_mhasalkar 94 Jun 19 05:25 p3.sh
4.0K -rwxrwxr-x 1 sneha_mhasalkar sneha_mhasalkar 9 Jun 19 06:32 practice.sh
4.0K -rwxrwxr-x 1 sneha_mhasalkar sneha_mhasalkar 470 Jun 19 11:48 q11.sh
4.0K -rwxrwxr-x 1 sneha_mhasalkar sneha_mhasalkar 158 Jun 19 08:09 que10.sh
4.0K -rwxrwxr-x 1 sneha_mhasalkar sneha_mhasalkar 104 Jun 19 11:35 que13.sh
4.0K -rwxrwxr-x 1 sneha_mhasalkar sneha_mhasalkar 307 Jun 19 06:27 que2.sh
4.0K -rwxrwxr-x 1 sneha_mhasalkar sneha_mhasalkar 122 Jun 19 06:37 que4.sh
4.0K -rwxrwxr-x 1 sneha_mhasalkar sneha_mhasalkar 158 Jun 19 06:56 que6.sh
```

3. List the contents of sub directory.


```

sneha_mhasalkar@cloudshell:~/day2$ ls -R
.:
array.sh  funfromanother.sh  practice.sh  que2.sh  que8.sh
code.sh   p1.sh              q11.sh       que4.sh  que9.sh
demo      p2.sh              que10.sh     que6.sh  ques11.sh
file.txt  p3.sh              que13.sh     que7.sh

./demo:
de  demo2  demo3  demo4.txt

./demo/de:

./demo/demo2:

./demo/demo3:
sneha_mhasalkar@cloudshell:~/day2$

```

15. For the given file contents write a shell script that prints Number of field delimiters:Line Number
 Note: in this file the field delimiter is comma(,)

```

[cloudera@quickstart ~]$ cat test.txt
abc,def,ghi
123,456,789
aaa,bbb,ccc
12,34
1,1,3
2
4,5,6,7
[cloudera@quickstart ~]$ 3:1,3:2,3:3,2:4,3:5

```

```
sneha_mhasalkar@cloudshell:~/day2$ cat test.txt
abc,dcf,sdf
123,345,456
aaa,bbb,ccc
12,34
1,1,3
2
4,5,6,7
sneha_mhasalkar@cloudshell:~/day2$ awk '{print gsub(/,/, "&") ":" NR}' test.txt
2:1
2:2
2:3
1:4
2:5
0:6
3:7
sneha_mhasalkar@cloudshell:~/day2$ █
```

Awk command is used to count the delimiter in each line and print

16. write a shell script to count total no of arguments. If total arguments are 2 then come out of the script else print the arguments.

```
sneha_mhasalkar@cloudshell:~/day2$ ./arg.sh hello world
total number of argumentss 2
the nubmer of arg are 2 exit
sneha_mhasalkar@cloudshell:~/day2$ ./arg.sh hello world shell
total number of argumentss 3
arguments are
hello
world
shell
sneha_mhasalkar@cloudshell:~/day2$ █
```

I have given the 2 input with argument 2 and 3